KONERU CHARAN

+91-9148527810 ⋄ Vijayawada, Andhra Pradesh , India
konerucharan04@gmail.com ⋄ GitHub

EDUCATION

Vellore Institute of Technology , Bhopal — **Cumulative GPA: 9.32**
B.Tech Computer Science Engineering
Expected 2027

Sri Sri Ravishankar Vidya Mandir , Bangalore — **Percentage: 90%**
Class XII, CBSE — 2023

MES Kishore Kendra Public School — **Percentage: 96.3%**
Class X, ICSE — 2021

PROJECTS

Deep Fake Detection — AI-generated Image Classification — Jan 2025 – Apr 2025
Machine Learning Development

- Designed and trained a **Convolutional Neural Network (CNN)** model to detect AI-generated or fake photos.
- Optimized model performance through dataset preprocessing, hyperparameter tuning, and evaluation metrics.
- Delivered model outputs and supported integration and deployment documentation for the Web team.
- **Learned:** Deep Learning, CNN-based Image Classification, Model Optimization, Cross-team Collaboration.

AI-based Room Generator — Style Transfer and Novel Room Synthesis — Apr 2025 – May 2025
Machine Learning + Web Integration

- Built two generative pipelines: (1) a **style-transfer model** that re-styles rooms while preserving existing furniture layout, and (2) a **generative model** that synthesizes entirely new room designs.
- Implemented models using **Tensorflow** with preprocessing, inference, and evaluation workflows.
- Built a Streamlit UI deployed via ngrok, enabling remote interactive uploads, previews, and downloads.
- **Learned:** Generative Modeling, Image-to-Image Translation, Model Inference Pipelines, Streamlit , ngrok.

Road Accident Detection — Image-based Accident Classification — Aug 2024 – Oct 2024
Machine Learning / Computer Vision

- Developed an image classification system to distinguish between accident and non-accident road scenes using a pre-trained CNN with transfer learning, improving accuracy while reducing training time.
- Implemented data preprocessing, augmentation, batching, and performance optimization using **TensorFlow/Keras**.
- Evaluated model performance using validation and testing datasets to ensure generalization.
- **Learned:** Transfer Learning, CNN Architectures, Image Dataset Pipelines, Model Evaluation Techniques.

TECHNICAL SKILLS

Programming: Python, C++, SQL, HTML, CSS
Frameworks & Libraries: TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn
Tools: Git, GitHub, Jupyter Notebook, VS Code

CERTIFICATIONS

- IBM Generative AI using watsonx
- Microsoft Azure Data Fundamentals (DP-900)
- NPTEL: Introduction to Machine Learning